Directed Cut Certificates through Fractional Forests and Shared Flows*

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Abstract

We study certificates for exact minimum cuts in directed graphs accessed by outgoing-cut queries. For arbitrary finite nonnegative real capacities, every n-vertex graph has a global-minimum-cut certificate with $O(n^{5/3})$ queries. The transcript forces both the optimum value and a specified optimum shore in every consistent nonnegative weighted graph, even when the competitor changes the support. The bound has no dependence on weight spread or numerical precision.

The proof combines linear-size certificates for feasible flow divergences, a cut-equivalent orientation, and an implicit union of fractional branching packings of total capacity $O(n\lambda)$. A separate certificate handles all shores of size at most k. The structural step deletes $O(n/k)$ vertices from any fractional forest so that every remaining induced set has density at most $1 - 1/k$. A rooted-cover inequality and a uniform-root decomposition give unit deletion progress at twice the target slack. Applied to the two branching packings, this yields a common source set; two shared flows and root-to-terminal flows cover the remaining cuts. A spread-sensitive refinement gives $O(n^{3/2}\sqrt{\log(2n)})$ certificates for unit-capacity directed inputs, with the same unrestricted competitor class. Directed terminal minimum cuts have $O(n)$-query certificates for both unit and arbitrary weights. These are certificate-existence results with unrestricted offline computation, not algorithms for discovering the witnesses.

1 Introduction and results

An outgoing-cut query reveals the total capacity leaving a chosen vertex set. It need not reveal any individual arc: a directed cycle and its reversal have the same outgoing-cut function [12, 19]. This makes a short minimum-cut certificate different from a sparse list of known arcs. The certificate must constrain all graphs compatible with its answers, including graphs whose orientations and support differ from those of the input.

A single terminal pair is amenable to a flow witness. An extreme signed flow has only a forest of unsaturated pairs; the remaining divergence is fixed by partial-star totals. The global problem needs a lower bound for every nontrivial shore. Certifying all $2(n-1)$ terminal pairs incident to a fixed root, the standard reduction of directed global minimum cut to terminal cuts [5], costs $O(n^2)$ queries. Our construction reduces the number of necessary terminals to $O(n/k)$, while separately certifying all small shores; this split into small and large shores follows Cen et al. [5, Section 2]. The key structural statement concerns fractional forests and is independent of the query model.

*This preliminary draft was written by Codex GPT-6 Astra; see the AI use statement just before the references. The main theorems have all been verified in Lean 4. Appendix A gives the correspondence between all numbered statements in the paper and their Lean counterparts, and notes where they differ. I have not yet reviewed the prose proofs here. Corrections are welcome.

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1.1 Graphs, queries, and certificate quantifiers

Let V be a labelled vertex set of size $n \geq 2$. A graph G is specified by finite capacities $a_{uv} \in \mathbb{R}_{\geq 0}$ for distinct $u, v \in V$, with $a_{vv} = 0$. Antiparallel arcs are allowed. Parallel arcs, if present, are aggregated into their ordered-pair capacity. A unit input has $a_{uv} \in \{0, 1\}$ on every ordered pair. Write

$$F_G(S) = \sum_{u \in S,\, v \notin S} a_{uv}, \qquad \lambda(G) = \min_{\varnothing \neq S \subsetneq V} F_G(S).$$

For distinct terminals s, t, put $\lambda_{st}(G) = \min_{s \in S,\, t \notin S} F_G(S)$. An oracle returns the exact real number $F_G(S)$ for any specified S.

Definition 1 (Witness certificate)**.** A global cut certificate for G is a finite family $\mathcal{Q} \subseteq 2^V$ and a specified optimum shore $S_* \in \mathcal{Q}$ such that

$$[a'_{uv} \geq 0 \text{ and } F_{G'}(Q) = F_G(Q) \text{ for all } Q \in \mathcal{Q}] \quad \Longrightarrow \quad F_{G'}(S_*) = \lambda(G') = \lambda(G).$$

Competitors G' range over all loopless nonnegative real-weighted graphs on V, with no restriction on their support or capacity size. For a terminal certificate, S_* must be an s–t shore and λ is replaced by λ_{st}. Let $C^{\mathrm{cut}}_{\mathrm{glob}}(G)$ and $C^{\mathrm{cut}}_{s,t}(G)$ denote the minimum numbers of queries in these respective definitions.

The certificate may be selected with complete knowledge of G. Only the number of outgoing-cut queries is charged. Witness description length, exact arithmetic, and offline optimization are unrestricted. Every constraint used on a competitor must nevertheless follow from counted original queries. In particular, an implicit auxiliary graph is not automatically supplied with a cut oracle.

There is no finite-bit or commensurability assumption on capacities. All equalities, comparisons, and feasibility assertions use exact real arithmetic. For rational inputs the same proofs use exact rational data; we make no bound on denominator size, bit operations, or witness encoding length. These statements do not provide an exact certificate from rounded or approximate oracle answers. All certificates are sound with certainty. Sampling appears only in offline existence arguments, from which a fixed successful witness is selected; no failure probability is part of the certificate guarantee.

1.2 Main statements

Theorem 2 (Arbitrary-weight global certificates)**.** *For every graph in Definition 1 and every integer $1 \leq k < n$, there is a global cut certificate of size*

$$O\left(nk^2 + nk\log^2(2n) + \frac{n^2}{k}\right). \tag{1}$$

In particular, $C^{\mathrm{cut}}_{\mathrm{glob}}(G) = O(n^{5/3})$. If $\lambda(G) = 0$, one query suffices.

The constants in this theorem are uniform over all inputs and all weight values. The $n^{5/3}$ bound follows by choosing $k = \min\{n - 1, \lceil n^{1/3} \rceil\}$; the logarithmic small-shore term is absorbed by the resulting power bound.

Theorem 3 (Terminal certificates)**.** *For every such graph and distinct s, t, $C^{\mathrm{cut}}_{s,t}(G) = O(n)$. If $\lambda_{st}(G) = 0$, one query suffices. The claim holds against the unrestricted competitor class even when the input is unit-capacity.*

The unit global refinement depends on the symmetrized pair weights, which are intrinsic to the cut function. For $\lambda > 0$, define

$$u_{\{v,w\}} = a_{vw} + a_{wv}, \qquad \nu = \min_{u_e>0} u_e, \quad U = \max_e u_e, \quad \kappa = U/\nu, \qquad q(t) = |\{e : 0 < u_e < t\}|.$$

Theorem 4 (Spread-sensitive refinement)**.** *If $\lambda > 0$, then*

$$C^{\mathrm{cut}}_{\mathrm{glob}}(G) = O\left(\min\left\{n^{5/3}, \inf_{0<t\le U}\left[q(t) + n^{3/2}\sqrt{\frac{U}{t}\log(2n)}\right]\right\}\right). \tag{2}$$

In particular, $C^{\mathrm{cut}}_{\mathrm{glob}}(G) = O(\min\{n^{5/3}, n^{3/2}\sqrt{\kappa\log(2n)}\})$. No bound on κ, on a competitor's weights, or on its support is assumed in the competitor class.

Corollary 5 (Unit-capacity global certificates)**.** *Every loopless unit-capacity directed input, with antiparallel arcs allowed, has a global cut certificate of size $O(n^{3/2}\sqrt{\log(2n)})$, sound against all consistent nonnegative real-weighted competitors.*

The corollary uses $u_e \in \{0, 1, 2\}$ and hence $\kappa \le 2$. It comes from the separate refinement in Theorem 4, not from substituting unit weights into the $n^{5/3}$ theorem. Together the results cover the four directed certificate settings:

Input capacities	Global minimum cut	Terminal minimum cut
Arbitrary nonnegative reals	$O(n^{5/3})$	$O(n)$
Unit capacities	$O(n^{3/2}\sqrt{\log(2n)})$	$O(n)$

All four entries certify an exact value and a specified optimum shore against the same unrestricted competitor class. They are consequences of shared structural statements, rather than four unrelated results.

1.3 Proof overview and relation to prior arguments

We first represent the outgoing-cut function by an undirected symmetrization and a vertex imbalance vector, as in [12, 1]. An extreme signed flow then gives an $O(n)$-query certificate for any feasible directed flow divergence. The proof explicitly transfers a witness to every consistent graph, including graphs with new positive pairs.

The next step makes a dense graph usable without reconstructing it. An extreme orientation leaves only $O(n)$ split pairs. Two extreme fractional branching packings retain total capacity at most $2(n-1)\lambda$ and have only $O(n)$ partially retained pairs. The resulting graph H is implicit: its degrees, pairs, and partial stars cost $O(1)$ original queries, whereas arbitrary cuts of H need not. Every consistent G' induces an H' cutwise dominated by G'.

For small shores, fix the heavy arcs and bound all remaining capacities by $O(\lambda/k^2)$ using grouped stars. Redirect the residual out-degree to a root in an explicit comparison graph. Any competitor's violating small shore is near-minimum in this one fixed graph, so, by the one-respecting argument of Cen et al. [5, Section 4], $O(k \log n)$ arborescences cover all possibilities. Centroid flow certificates, following [5, Section 5], handle their connected shores in $O(n \log n)$ queries per tree. This costs $O(nk^2 + nk\log^2 n)$ queries altogether.

For the other shores, normalize the underlying vectors of the two branching packings to fractional forests. We prove that deleting $O(n/k)$ vertices makes every remaining induced density at most $1 - 1/k$. The union of the two deletion sets and their common root is then a source set R: all

outside vertices can send λ/k units to R, and can separately receive that amount from R. Two certified flows cover large shores not splitting R; $O(|R|)$ terminal flows cover cuts that do split it. This gives the last term in (1).

The flow-certificate argument originates in the companion manuscript *Finding and Certifying Weighted Minimum Cuts with Cut Queries* [21]; we give the directed proof here. An earlier unit-capacity certificate combines branching packings with two shared star flows, but its large-value construction assumes competitors have arc capacities at most one. Implicit pruning removes that restriction, and the fractional-forest argument extends the construction to real weights. This paper organizes the proofs around the weight-independent theorem and proves the unit bound as a refinement. Section 10 records the earlier optimum-dependent bound with its precise, narrower competitor promise.

The external combinatorial input is Edmonds' integral arborescence packing theorem [7]; Gao and Yang [10, Theorem 1.1] state the prescribed-root form used here. Its real-capacity version, the exact duality between fractional arborescence packings and minimum rooted cuts, is also used by Cen et al. [5, Section 4], who cite Edmonds and Gabow [9]; we include a short compactness derivation. The centroid step follows the one-respecting arborescence framework of Cen et al. [5, Section 5], with its query certificate and competitor-transfer proof included below. We also use the max-flow/min-cut theorem of Ford and Fulkerson [8] for finite real capacities. No spectral sparsifier or certificate-discovery procedure is needed for these results.

Related work. Rubinstein, Schramm, and Weinberg [27] introduced cut queries to study exact optimization from aggregate graph information, motivated by value-oracle submodular function minimization, and asked whether weighted or directed cuts can be computed with $o(n^2)$ queries. Randomized algorithms find undirected global minimum cuts with $\widetilde{O}(n)$ queries in weighted graphs [25] and $O(n)$ queries in simple graphs [2]. A directed outgoing-cut function is submodular, and a global minimum cut is a nontrivial minimizer. General submodular minimization uses $\widetilde{O}(n^2)$ value queries [23, 16], and the best known deterministic lower bound is $\Omega(n \log n)$ [6], improving the $2n$ bound of Graur et al. [12]. Graur et al. also prove a $\binom{n}{2}$ lower bound for a nontrivial minimizer of a general asymmetric submodular function; their hard functions differ on a single set, so the bound applies to certificates as well. For directed cut functions, Theorem 2 gives polynomially smaller certificates.

Graur et al. [12] introduced the cut dimension as a lower bound for deterministic cut-query algorithms, and showed that for s–t cuts it is at most $n+1$, even in directed weighted graphs. For undirected global minimum cuts, Lee, Li, Santha, and Zhang [22] showed that the cut dimension is at most $2n-3$ and introduced the ℓ_1-approximate cut dimension, which Auza and Lee [3] showed characterizes, up to an additive one, the linear-query certificate complexity of minimum cut. The companion manuscript [21] gives $O(n \log^2(2n))$-query cut certificates for undirected weighted global minimum cuts.

Algorithms with directed cut queries are recent. Khanna, Putterman, and Song [19] give $O(n \log^3 n)$-query reachability and topological sorting in directed acyclic graphs, and cycle detection in arbitrary digraphs; Bals, Tinca, and Nazari [4] give a deterministic $O(n\sqrt{n \log n})$-query single-source reachability algorithm for directed acyclic graphs. An anonymous note listed in Yang P. Liu's repository [1, 24] gives randomized $\widetilde{O}(n^{3/2})$-query reachability in simple digraphs, a problem posed by Nanongkai [26] and highlighted in the open-problem discussion of [17]; Nanongkai's page reports that a few groups have reached roughly $n^{3/2}$ queries. Ismaili Alaoui, Lachish, and Mande [15] show that finding a king in a tournament takes $\Theta(n)$ deterministic outgoing-cut queries. The present results concern certificates, not algorithms; see Section 8.

2 Cut identities and flow certificates

A feasible directed flow is a vector q with $0 \le q_{uv} \le a_{uv}$; its net-outflow is $d_v = \sum_w (q_{vw} - q_{wv})$. The in- and out-degrees are $d_G^-(v) = \sum_w a_{wv}$ and $d_G^+(v) = \sum_w a_{vw}$.

Fix a canonical orientation of unordered pairs, with incidence matrix D using $+1$ at the tail. Let $\mathcal{U}$ be the undirected symmetrization. Write

$$u_{\{v,w\}} = a_{vw} + a_{wv}, \qquad \beta_v = d_G^+(v) - d_G^-(v).$$

The singleton and complementary-singleton queries learn β and both degree vectors using at most $2n$ queries. For every shore,

$$2F_G(S) = c_{\mathcal{U}}(S) + \beta(S). \tag{3}$$

For $v \notin T$ a partial star of the symmetrization is

$$u(v,T) = F_G(\{v\}) + F_G(T) - F_G(T \cup \{v\}). \tag{4}$$

Graur et al. [12, Appendix B] learn the degree vectors and symmetrized pair capacities in this way, and observe that cut queries determine a weighted digraph only up to adding and deleting directed cycles. Identities (3) and (4) are the cut algebra of [1, Lemma 2.2]; (4) is also the symmetric crossing identity of [19, Lemma 2.1]. Ghosh et al. [11] show more generally that the skeleton and indegrees of a digraph determine every property invariant under cycle reversal.

In particular, singleton and pair queries recover all u_e in $O(n^2)$ queries. Together with β, this determines the entire outgoing-cut function by (3), even though individual directed arcs are not determined [4, 26]; the graph reconstruction technique of Grebinski and Kucherov [13] saves a logarithmic factor [4]. Offline optimization gives a quadratic certificate fallback.

Lemma 6 (Arbitrary real flow divergences)**.** *If q is a feasible directed flow in G, with net-outflow vector d, then $O(n)$ outgoing-cut queries certify*

$$F_{G'}(S) \ge d(S) \qquad \text{for all } S$$

in every consistent nonnegative weighted graph G'.

Proof. For a canonical pair $v \to w$, put

$$z_{vw} = a_{vw} - a_{wv} - 2(q_{vw} - q_{wv}).$$

Then $Dz = \beta - 2d$ and $|z_e| \le u_e$. Choose an extreme point of the compact feasible polytope $\{z : Dz = \beta - 2d,\ -u \le z \le u\}$. Its unsaturated positive-capacity edges form a forest: a cycle would admit a small circulation in either direction. Query these at most $n-1$ pair capacities. On every other originally positive edge the signed flow is saturated with a prescribed sign. Two partial stars per vertex fix the signed saturated contribution to its divergence. Originally zero-capacity pairs carry zero.

In a consistent graph, keep the forest flows fixed, use the new capacities with the prescribed signs on saturated edges, and put zero on originally zero-capacity pairs. This gives a feasible signed flow z' with $Dz' = \beta - 2d$. Hence

$$2(F_{G'}(S) - d(S)) = c_{\mathcal{U}'}(S) + (\beta - 2d)(S) \ge 0.$$

All the required pair and star measurements are implemented using (4). This is the weighted version of the unit-capacity star-flow construction of Proposition 28, and the directed form of the flow certificate of [21, Lemma 2], which obtains the forest from a minimum-cost flow with generic costs rather than from an extreme point. □

Lemma 7 (Sparse weighted domination)**.** *For any explicit capacitated directed subgraph $0 \le H \le G$ with s positive arcs, $O(n+s)$ queries certify $F_{G'}(S) \ge F_H(S)$ for every S, against all nonnegative weighted competitors.*

Proof. Let h be the symmetrization of H and let β^H be its degree imbalance. Query the symmetrized capacity of every pair supporting H, as well as the original degree vectors. Then $c' = u' - h \ge 0$ in every consistent graph. The actual residual graph $G - H$ supplies a signed flow in capacities $c = u - h$ with divergence $\beta - \beta^H$. Certify this divergence by the extreme-flow construction in Lemma 6; a residual partial star is an original partial star minus the known contribution of h. There are at most $n-1$ further forest-pair measurements and $O(n)$ star measurements. The resulting inequality $c_{c'}(S) \ge |(\beta - \beta^H)(S)|$ gives

$$2(F_{G'}(S) - F_H(S)) = c_{c'}(S) + (\beta - \beta^H)(S) \ge 0.$$

This proves cutwise domination without claiming that directed arcs of H are individually identifiable. □

Proof of Theorem 3. Choose a maximum feasible s–t flow in the actual graph, of value λ_{st}. Max-flow/min-cut [8] holds for finite real capacities: a maximum exists by compactness, and residual reachability supplies a matching cut. Its divergence vector is $d = \lambda_{st}(\mathbf{1}_s - \mathbf{1}_t)$. Lemma 6 certifies $F_{G'}(S) \ge \lambda_{st}$ for every s–t shore. Querying one actual optimum s–t shore S_* gives the reverse inequality and keeps that shore optimal in every competitor. The total cost is $O(n)$. If $\lambda_{st} = 0$, only the optimum-shore query is needed, because all competitors have nonnegative cuts. □

The proof of Lemma 6 uses only degree vectors, symmetrized pair capacities, and symmetrized partial stars. Thus it also applies to an auxiliary graph when precisely these measurements are available. It does not require arbitrary cut queries to that auxiliary graph. This observation will be used at every later compilation step. Section 9 gives the alternative reduction from directed terminal certificates to undirected terminal certificates.

3 Fractional branchings and implicit pruning

From this section onward, write $\lambda = \lambda(G)$ and assume $\lambda > 0$ whenever a construction divides by λ. A zero global optimum is certified by querying its shore and using nonnegativity.

An out-arborescence rooted at r is a spanning tree whose arcs point away from r; an in-arborescence points toward r. A fractional packing of value α is a finite nonnegative linear combination of such trees, with coefficients summing to α and arc loads bounded by the available capacities.

Lemma 8 (Real-capacity branching packing)**.** *If every nonempty $S \subseteq V \setminus \{r\}$ has incoming capacity at least $\alpha \ge 0$, then the graph has a fractional packing of out-arborescences rooted at r of value α. The analogous outgoing-cut condition gives an in-arborescence packing.*

Proof. The zero case is immediate. For a positive integer j, replace each capacity a_e by the integer $\lfloor j a_e \rfloor$. If there are m potential arcs, every relevant incoming cut now has capacity at least $j\alpha - m$. Edmonds' theorem [7, 10] packs $\max\{0, \lfloor j\alpha - m \rfloor\}$ arc-disjoint rooted out-arborescences in this integer-capacity multigraph. Divide each tree coefficient by j. There are only finitely many rooted arborescence types on V, their coefficients are nonnegative and bounded, and their sum tends to α. A convergent subsequence therefore gives a packing of value α whose load on each arc is at most a_e. Reversing all arcs proves the in-arborescence statement. □

Lemma 9 (Sparse weighted spanning certificate)**.** *Assume $\lambda > 0$, and let μ be the smallest positive arc capacity of G. There exists $0 \le H \le G$ with global mincut at least λ and with at most*

$$2(n-1)\lambda/\mu + 4n - 6$$

positive arcs.

Proof. Fix a root r. First consider the polytope

$$P = \{x : 0 \le x \le a, \quad x(\delta^-(S)) \ge \lambda \ (\varnothing \ne S \subseteq V \setminus \{r\})\}.$$

Lemma 8 gives a point of P with total capacity $\lambda(n-1)$. Conversely, the singleton constraints give total capacity at least $\lambda(n-1)$. Thus this is the optimum of minimizing $\sum_e x_e$ over P. Any optimal point itself equals the load vector of a packing of value λ: apply Lemma 8 to that point, and note that domination together with equal total capacity forces equality on every arc.

Choose an extreme optimal point x. We claim that at most $2n-3$ arcs satisfy $0 < x_e < a_e$. Let K be this set of free arcs. The tight cut constraints must span $\mathbb{R}^K$: otherwise a nonzero perturbation on K would preserve them, and sufficiently small perturbations in both directions would preserve all remaining inequalities and bounds, contradicting extremality.

The tight cut rows restricted to K have a laminar spanning family, by the uncrossing argument for rooted-cut linear programs [20, Section 6.1]. Here are the details. If tight sets A, B intersect and neither contains the other, submodularity and feasibility force $A \cap B, A \cup B$ to be tight. Moreover,

$$x(\delta^-(A)) + x(\delta^-(B)) - x(\delta^-(A \cap B)) - x(\delta^-(A \cup B)) = x(A \setminus B, B \setminus A) + x(B \setminus A, A \setminus B) = 0.$$

Both crossing arc sets therefore vanish on the positive support of x, so the same cut-row identity holds on K. Take an inclusion-maximal linearly independent laminar family $\mathcal{L}$ of tight rows. If a tight set S has a row outside its span, choose such S crossing as few members of $\mathcal{L}$ as possible. If it crosses $A \in \mathcal{L}$, the row identity implies that $S \cap A$ or $S \cup A$ is also outside the span. Each of these crosses strictly fewer members of $\mathcal{L}$: laminarity ensures no new crossing is created, and the crossing with A disappears. This is a contradiction. If S crosses no member, it can instead be added to $\mathcal{L}$, also a contradiction. Thus $\mathcal{L}$ spans all tight rows. A laminar family of nonempty subsets of the $(n-1)$-element set $V \setminus \{r\}$ has size at most $2n-3$, proving the claim.

Every remaining positive arc of x is saturated and has capacity $a_e \ge \mu$. Since $\sum_e x_e = \lambda(n-1)$, there are at most $\lambda(n-1)/\mu$ such arcs. Repeat the construction in the reversed graph, obtaining a capacity vector y for in-arborescences in the original graph. Let $h_e = \max\{x_e, y_e\}$. Then $h \le a$ and the support bound follows. A nontrivial shore containing r has outgoing x-capacity at least λ, and one avoiding r has outgoing y-capacity at least λ. Hence H has the required global lower bound. □

Corollary 10. *With μ as in Lemma 9, there is a certificate of size $O(n + n\lambda/\mu)$, sound against all nonnegative weighted competitors.*

Proof. Apply Lemma 7 to H and query an optimum shore. □

The fractional extreme-point step matters here. Merely replacing each positive weighted arc by a unit arc, or rounding all capacities down, does not preserve the exact value λ.

Lemma 11 (A cut-equivalent orientation with few split pairs)**.** *Using $O(n)$ queries, one can certify an implicit representation G° with the same outgoing-cut function as G. Outside at most $n-1$ exceptional unordered pairs, every originally positive pair has all its capacity u_e in one prescribed direction. The capacities on exceptional pairs are fixed by the queries. For every consistent competitor G', the same description gives a nonnegative $G^{\circ\prime}$ with exactly the cut function of G'.*

Proof. The signed vector $a_{vw} - a_{wv}$ is feasible in capacities u with divergence β. Choose an extreme signed flow z with $Dz = \beta$ and $|z_e| \le u_e$. Its unsaturated edges form a forest. Query their capacities and the saturated partial-star totals, as in Lemma 6, as well as the original degree vectors. For a canonical pair $v \to w$, define

$$a^\circ_{vw} = (u_e + z_e)/2, \qquad a^\circ_{wv} = (u_e - z_e)/2.$$

These are nonnegative, have symmetrization u and imbalance β, and therefore have the same cut function by (3). On a nonexceptional positive pair, $z_e = \pm u_e$, so there is only one positive direction.

For a consistent competitor, keep z fixed on forest edges, use the new capacities with the prescribed signs on originally positive saturated pairs, and put $z'_e = 0$ on originally zero pairs. The certificates give $Dz' = \beta$ and $|z'_e| \le u'_e$. The same formulas define $G^{\circ\prime}$. Originally zero pairs may become antiparallel pairs of capacity $u'_e/2$ in each direction; they need not remain zero. This causes no difficulty because the graph retained below ignores them. □

Lemma 12 (An implicit graph of small total capacity)**.** *After $O(n)$ queries there is a description of a graph H with global mincut λ and total arc capacity at most $2(n-1)\lambda$ such that:*

(i) on $O(n)$ exceptional pairs, its two directed capacities are explicit constants;

(ii) on each other retained pair, it has one prescribed direction with capacity equal to the original unknown u_e;

(iii) on all other pairs its capacities are defined to be zero.

Every consistent competitor induces a graph H' with $F_{G'}(S) \ge F_{H'}(S)$ for every S. Each degree, pair-capacity, or directed or symmetrized partial-star measurement of H' costs $O(1)$ outgoing-cut queries to G'. Furthermore, if M_H is the largest arc capacity of H, then

$$M_H \le \min\{\lambda, U\}, \qquad |\operatorname{supp}(H)| \le q(t) + \frac{2(n-1)\lambda}{t} + O(n) \quad (t > 0).$$

Proof. Apply Lemma 11. In the actual graph G°, choose the two extreme minimum-total-capacity rooted packings x, y from the proof of Lemma 9, and set $h_e = \max\{x_e, y_e\}$. Both packings have total capacity $\lambda(n-1)$, so H has total capacity at most $2\lambda(n-1)$. Its global mincut is at least λ, and $H \le G^\circ$ gives equality by evaluating an original optimum shore.

Each packing has at most $2n-3$ unsaturated positive arcs. Declare their unordered pairs, together with the orientation forest, exceptional, and query their original symmetrized capacities. There are $O(n)$ exceptional pairs. Outside them, every positive retained arc is a saturated arc of at least one packing, hence is exactly the single arc of G° on that pair. Its capacity is u_e with a prescribed orientation. Retain fixed capacities h on exceptional pairs.

In a consistent graph, define H' using the same masks and constants. On ordinary retained pairs it takes the full prescribed arc of $G^{\circ\prime}$; on exceptional pairs the containing capacities are fixed; elsewhere it takes zero. Thus $H' \le G^{\circ\prime}$ arcwise, proving cutwise domination. An originally absent edge that appears in a competitor is simply ignored by H'.

For a partial star of H', query the corresponding subset of ordinary retained neighbors in the original symmetrization, and add the known exceptional contribution. For an outgoing or incoming degree, use just the retained neighbors with the appropriate prescribed direction, plus the fixed arcs. Pair measurements are equally direct. An arbitrary cut of H' need not have such a constant-cost simulation.

In the out-arborescence packing, every nonroot vertex has total in-capacity exactly λ and the root has in-capacity zero; in the in-arborescence packing the analogous statement holds for out-capacity. Consequently every arc of either packing is at most λ. Also $h \le a^\circ$ and every $a_e^\circ \le U$, proving the bound on M_H.

An ordinary retained arc of capacity less than t corresponds to one of the $q(t)$ original unordered pairs. All other ordinary retained arcs contribute at least t to the total capacity $2(n-1)\lambda$. The exceptional pairs contribute only $O(n)$ to the support count, irrespective of how tiny their fixed capacities are. □

Lemma 13 (Cap verification charged to total capacity)**.** *For the implicit graph H of Lemma 12 and any $B \ge 2M_H$, $O(n + n\lambda/B)$ original queries ensure that every induced graph H' has arc capacities at most B.*

Proof. At each vertex, group its ordinary retained neighbors into bins whose actual symmetrized capacity totals are at most B. Each item has weight at most $M_H \le B/2$. With next-fit grouping, every bin except the last has weight greater than $B/2$, so at most $1 + 2d_v/B$ bins suffice, where d_v is the ordinary retained star capacity. Query their totals using Lemma 12. Nonnegativity forces each constituent capacity in a competitor to be at most B. The exceptional directed capacities are fixed and at most M_H. Finally, $\sum_v d_v \le 2\sum_e h_e \le 4(n-1)\lambda$, giving the query bound. The bins depend on the input, as certificate selection permits. □

4 A weight-independent certificate for small shores

We first certify every cut with a small shore. The comparison graph used in this argument is fixed by the transcript, so the tree family covers potential violations in all competitors simultaneously.

Theorem 14 (All small shores, with unrestricted competitors)**.** *For every integer $1 \le k < n$, there is a certificate of size*

$$O(nk^2 + nk\log^2(2n))$$

forcing

$$F_{G'}(S) \ge \lambda \quad \textit{whenever} \quad 0 < \min\{|S|, |V \setminus S|\} \le k$$

in every consistent nonnegative weighted competitor. There is no dependence on a capacity ratio or numerical precision.

Lemma 15 (Certifying connected shores of a prescribed tree)**.** *Let H have global mincut at least λ, and suppose its directed partial stars and symmetrized pair capacities are available at constant query cost. For any prescribed tree T on its vertices and root r, $O(n\log(2n))$ measurements certify*

$$F_{H'}(S) \ge \lambda$$

for every nonempty T-connected set S avoiding r. Neither the tree edges nor their orientations need belong to H.

Proof. Remove r from T, then recursively remove a centroid from each remaining tree component. At a fixed depth the components W are disjoint, and the depth is $O(\log(2n))$. For a component W with centroid c, fix, for each $v \in W$, its incoming and outgoing H-capacity to $V \setminus W$ by directed partial-star measurements. Form the local graph on $W \cup \{z\}$ by contracting $V \setminus W$ into z: its internal arcs remain implicit, and its boundary arcs are the measured constants. Every local cut containing c and avoiding z is a nontrivial outgoing cut of H, so a c–z flow of value λ exists. Certify

it by Lemma 6, at cost $O(|W|+1)$. All measurements in this local certificate compile to partial stars or pair measurements inside W, plus known boundary constants.

If S is connected in T and avoids r, consider the first centroid belonging to S. Until that centroid is removed, S lies entirely inside its component W: a connected set avoiding all previously removed vertices cannot meet two components. Thus the corresponding local flow certifies $F_{H'}(S) \geq \lambda$. Summing over the disjoint components at each depth proves the bound. This is the certificate version of the centroid reduction for one-respecting arborescences in [5, Section 5]. □

Proof of Theorem 14. Use Lemma 12 to obtain the implicit graph H of total capacity at most $2(n-1)\lambda$, and fix all its degrees. The case $\lambda = 0$ is immediate, so assume $\lambda > 0$. Set

$$\tau = \frac{\lambda}{16k^2}.$$

Let K consist of all fixed exceptional arcs and every ordinary retained arc with actual capacity at least τ. There are $O(n + n\lambda/\tau) = O(nk^2)$ such arcs, and their capacities can be fixed individually. Write $H' = K + L'$ for every induced competitor; the actual remainder L has all arc capacities below τ. Group only these remaining ordinary pairs into partial-star bins of total at most 2τ. The same next-fit argument as in Lemma 13 uses $O(n + n\lambda/\tau) = O(nk^2)$ queries and forces

$$0 \leq L'_{vw} \leq 2\tau$$

on every remaining arc. The other residual arcs are defined to be zero. The residual in- and out-degrees are fixed because those of H and all capacities of K are fixed.

Fix a root r. Construct the *explicit comparison graph* A from K by adding an arc $v \to r$ of capacity $d^+_L(v)$ for every $v \neq r$, aggregating parallel arcs. For every S avoiding r and every consistent competitor,

$$F_A(S) = F_{H'}(S) + L'(S, S). \tag{5}$$

Here $L'(S, S)$ counts the directed arcs with both endpoints in S. In particular, the actual graph gives $F_A(S) \geq \lambda$ for every nonempty S avoiding r. Moreover, if $|S| \leq k$ and $F_{H'}(S) < \lambda$, then

$$F_A(S) < \lambda + 2\tau k(k-1) \leq \frac{9}{8}\lambda.$$

The comparison graph is the same in all competitors; it is not claimed to be a subgraph of any of them.

Pack in-arborescences rooted at r in A with total weight λ, using the real-capacity form of the branching theorem. For a random tree T drawn from this packing and a nonempty $S \subseteq V \setminus \{r\}$, let $X_T(S)$ count its arcs leaving S. Then

$$X_T(S) \geq 1, \qquad \mathbb{E}X_T(S) \leq F_A(S)/\lambda.$$

Consequently each set in the fixed family

$$\mathcal{F} = \{S \subseteq V \setminus \{r\} : 1 \leq |S| \leq k,\ F_A(S) < 9\lambda/8\}$$

is one-respected with probability greater than $7/8$. There are at most n^k sets in this family. Hence $O(k \log(2n))$ trees suffice to one-respect every member. Equivalently, choose trees greedily from the packing: some tree covers more than $7/8$ of the remaining sets, so the number remaining decreases geometrically. This is the one-respecting argument of Cen et al. [5, Section 4], the directed analogue

of Karger's tree packing [18], applied to a fixed comparison graph. All this is offline certificate selection.

A nonempty set avoiding the root and having exactly one outgoing arc in an in-arborescence is connected in the underlying tree. Apply Lemma 15 to every selected tree in H. The resulting $O(nk\log^2(2n))$ measurements exclude every consistent competitor's small violating shore that avoids r, by (5) and the fixed cover of $\mathcal{F}$. Thus the tree-cover step holds simultaneously for all competitors; it is not merely a successful randomized search on the original input.

To cover small shores containing r, choose k further distinct vertices $r_1,\dots,r_k$ and certify a value-λ flow from r to each r_i. Every such shore misses at least one r_i. This costs $O(nk)$. Finally repeat the comparison-graph and tree construction in the reversal of H, using its fixed residual in-degrees, and certify the reverse terminal flows. This covers every shore with small complement as well. Cutwise domination transfers all lower bounds from H' to G'. □

5 Density deletion in fractional forests

A nonnegative vector x on the unordered pairs of V is a *fractional forest* if

$$x(E(S)) \le |S| - 1 \qquad (\varnothing \ne S \subseteq V), \tag{6}$$

where each unordered pair is counted once. Convex combinations of spanning-tree incidence vectors are examples. The aim of this section is to delete $O(n/k)$ vertices so that the stronger density inequality $x(E(S)) \le (1-1/k)|S|$ holds on every remaining induced set. A separator for a single sampled tree does not provide this simultaneous statement for a fractional forest.

For $0 < \epsilon < 1$, define

$$g(S) = x(E(S)) - (1-\epsilon)|S|, \qquad D(W) = \max_{S\subseteq W} g(S), \qquad \Delta_v(W) = D(W) - D(W\setminus\{v\}).$$

The empty set is allowed, so $D(W) \ge 0$. The function g is supermodular because all pair weights are nonnegative. Its monotone closure D is also supermodular: if $A_0 \subseteq A$ and $B_0 \subseteq B$ attain the two maxima, then

$$D(A) + D(B) = g(A_0) + g(B_0) \le g(A_0 \cup B_0) + g(A_0 \cap B_0) \le D(A\cup B) + D(A\cap B).$$

Taking $A = W\setminus\{v\}$ and $B = K \subseteq W$ with $v \in K$ gives $\Delta_v(W) \ge \Delta_v(K)$.

Lemma 16 (Localization to maximum density)**.** *Suppose $D(W) > 0$, and choose a nonempty $K \subseteq W$ maximizing $\rho = x(E(K))/|K|$. Put $h = \rho - (1-\epsilon) > 0$. Then $D(K) = h|K|$ and*

$$\frac{D(K)}{|K|} \ge \frac{D(W)}{|W|}, \qquad \Delta_v(W) \ge \Delta_v(K) \quad (v \in K).$$

For every $T \subseteq K$, its loss inside K satisfies

$$\ell_K(T) := g(K) - g(K\setminus T) \ge h|T|.$$

Proof. For $S \subseteq K$, maximum density gives $g(S) \le h|S| \le h|K| = g(K)$, proving $D(K) = g(K)$. Applying the same density bound to a maximizer for $D(W)$ gives $D(W) \le h|W|$. Supermodularity of D gives $\Delta_v(W) \ge \Delta_v(K)$. Finally, $g(K\setminus T) \le h(|K| - |T|)$ proves the loss estimate. □

Lemma 17 (Minimal covers in a rooted forest)**.** *Let F be a forest on a nonempty vertex set K, with each component oriented toward its root, and let R_F be its root set. If $T_1, \dots, T_m$ is a cover of K from which no member can be removed, then*

$$\sum_{i=1}^{m} \Big(|\delta_F^-(T_i)| + |T_i \cap R_F| \Big) \geq m. \tag{7}$$

Consequently, for every $\gamma > 0$,

$$\sum_{i=1}^{m} \Big(|\delta_F^-(T_i)| + \gamma |T_i \cap R_F| \Big) \geq m \min\{1, \gamma\}.$$

Here $\delta_F^-(T)$ consists of arcs entering T.

Proof. Label a vertex v by the nonempty set $I(v) = \{i : v \in T_i\}$. Minimality gives a private witness v_i with $I(v_i) = \{i\}$ for every i. Writing $p(v)$ for the parent of a nonroot vertex, the left side of (7) equals

$$L = \sum_{r \in R_F} |I(r)| + \sum_{v \notin R_F} |I(p(v)) \setminus I(v)|.$$

Process the vertices of each tree from the root toward its leaves, shrinking every label to a singleton. At a root keep any label. At a nonroot vertex whose parent's label is now $\{i\}$, keep i if available, and otherwise keep any of the vertex's labels. This leaves the term on its arc to the parent unchanged. Shrinking a vertex's label cannot increase a term on an arc entering it from a child; at a root it also decreases the root term. Thus L never increases. Private singleton witnesses keep their labels, so all m labels still occur at the end.

For singleton labels, L is the number of roots plus the number of edges whose endpoint labels differ. Removing those edges from the forest shows that this is exactly the number of monochromatic connected components. It is at least the number m of labels. This proves (7). The weighted assertion follows by multiplying the unweighted left side by $\min\{1, \gamma\}$ and using nonnegativity of both counts. □

Corollary 18 (Concentration of fractional roots)**.** *Let y be the expected arc vector of a distribution of rooted forests on a nonempty set K, oriented toward their roots, and let r_v be the probability that v is a root. For every $\gamma > 0$ there is a vertex v such that*

$$y(\delta^-(T)) + \gamma r(T) \geq \min\{1, \gamma\} \qquad (v \in T \subseteq K).$$

Proof. Otherwise the sets with cost strictly less than $\min\{1, \gamma\}$ cover K. Choose a subcover with no redundant member, of size m. Taking expectations in Lemma 17 gives a total cost at least $m \min\{1, \gamma\}$, contradicting the strict upper bound on each member's cost. □

Lemma 19 (Uniform roots on a densest induced set)**.** *Suppose $x \geq 0$ on the unordered pairs of a nonempty set K satisfies the fractional-forest inequalities $x(E(S)) \leq |S| - 1$ for every nonempty $S \subseteq K$. Put $\rho = x(E(K))/|K|$, and suppose $x(E(S)) \leq \rho |S|$ for all $S \subseteq K$. There is a distribution of rooted forests on K, oriented toward their roots, whose expected arc vector y satisfies*

$$y_{uv} + y_{vu} = x_{\{u,v\}}, \qquad d_y^+(v) = \rho, \qquad \Pr[v \text{ is a root}] = 1 - \rho \quad (v \in K).$$

Proof. First split each edge capacity between its two endpoints, so that every vertex receives total capacity ρ. This is a bipartite flow problem: an edge supplies x_e, can send it to either endpoint, and each vertex can receive at most ρ. For any collection of edges with endpoint set S, its total supply is at most $x(E(S)) \le \rho|S|$. These are the max-flow/min-cut conditions for sending all supplies. Since their sum is $\rho|K|$, every vertex capacity is saturated. Interpret the allocation to an endpoint as arc capacity directed from that endpoint; this gives y with $d_y^+(v) = \rho$. This step is the fractional form of Hakimi's orientation theorem [14].

Put $\delta = 1 - \rho > 0$, where positivity follows from the forest inequality for K. Add a vertex q and an arc $v \to q$ of capacity δ at every $v \in K$. Each such vertex now has out-capacity one. For every nonempty $S \subseteq K$, its outgoing capacity in the augmented graph is

$$y(\delta^+(S)) + \delta|S| = |S| - x(E(S)) \ge 1.$$

Fractional in-arborescence packing rooted at q therefore gives a packing of total value one dominated by these capacities. For real capacities this follows from the rounding and limit argument in Lemma 8. There are finitely many arborescences, so the limit is a convex combination of their incidence vectors.

Every arborescence has one outgoing arc at every $v \in K$. The packing and capacity vector consequently have the same out-degree one at every such vertex. Domination then forces equality on every arc. Removing q from each arborescence leaves a rooted forest. A vertex is a root precisely when its arc to q was present, so every root marginal is exactly δ, as required. □

Theorem 20 (Deletion progress with arbitrary overlap)**.** *Let x be a fractional forest, $0 < \epsilon < 1$, and $D(W) > 0$, with g, D, Δ as in Section 5. Choose a densest nonempty $K \subseteq W$, put*

$$\rho = \frac{x(E(K))}{|K|}, \qquad h = \rho - (1-\epsilon) > 0, \qquad \delta = 1 - \rho > 0.$$

Some $v \in K$ satisfies

$$\Delta_v(W) \ge \Delta_v(K) \ge \min\{1, h/\delta\} \ge \frac{h}{\epsilon}.$$

In particular,

$$\max_{v \in W} \Delta_v(W) \ge \frac{D(W)}{\epsilon|W|} \ge \frac{D(W)}{\epsilon n}. \tag{8}$$

Proof. Lemma 16 gives $D(K) = g(K)$, $h \ge D(W)/|W|$, and $\Delta_v(W) \ge \Delta_v(K)$. Apply Lemma 19 on K. Since $d_y^+(v) = \rho$, the loss of deleting $T \subseteq K$ is

$$\begin{aligned} \ell_K(T) &= x(E(T)) + x(T, K \setminus T) - (1-\epsilon)|T| \\ &= y(\delta^-(T)) + h|T|. \end{aligned}$$

The root marginal is $r_v = \delta$. Apply Corollary 18 with $\gamma = h/\delta$. It supplies a vertex v for which every $T \ni v$ has $\ell_K(T) \ge \min\{1, h/\delta\}$. As $D(K) = g(K)$,

$$\Delta_v(K) = \min_{v \in T \subseteq K} \ell_K(T).$$

Finally $h + \delta = \epsilon$, so $\min\{1, h/\delta\} \ge h/(h+\delta) = h/\epsilon$. This proves all assertions. □

Corollary 21 (Structural density deletion without a logarithm)**.** *For every fractional forest x on n vertices and integer $1 \le k \le n$, there is $Z \subseteq V$ of size $O(n/k)$ such that*

$$x(E(S)) \le (1 - 1/k)|S| \qquad (S \subseteq V \setminus Z).$$

More precisely, for $2 \le k \le n$, one can take

$$|Z| \le \left\lceil \frac{2n-1}{k} \right\rceil - 2 \le 2(\lceil n/k \rceil - 1).$$

For $k = 1$, $n-1$ deletions suffice. No weight-precision assumption is needed.

Proof. Suppose $k \ge 3$ and set $\epsilon = 2/k$. Stop as soon as every remaining induced set has density at most $1 - 1/k$. While this fails, a densest nonempty $K \subseteq W$ has $\rho > 1 - 1/k$. Thus the parameters in Theorem 20 satisfy

$$0 < \delta = 1 - \rho < 1/k, \qquad h = 2/k - \delta > \delta.$$

The theorem supplies a vertex with

$$D(W) - D(W \setminus \{v\}) \ge \min\{1, h/\delta\} = 1.$$

Deleting a vertex of maximum gain has the same guarantee. Initially the forest inequalities give $D(V) \le 2n/k - 1$. Moreover, any violating set S has $|S| \ge k+1$, since $x(E(S)) \le |S| - 1$. Therefore, while a violation remains,

$$D(W) \ge x(E(S)) - (1 - 2/k)|S| > |S|/k \ge 1 + 1/k.$$

After $q = \lceil (2n-1)/k \rceil - 2$ deletions the unit-progress bound gives $D(W) \le 2n/k - 1 - q \le 1 + 1/k$, so no violation can remain. The integer q is nonnegative because $n \ge k$. For $k = 2$, retain any two vertices: their pair weight is at most one, and the stated bound is $n - 2$. For $k = 1$, retain one vertex. Finally, $\lceil (2n-1)/k \rceil \le 2\lceil n/k \rceil$ proves the displayed comparison with the sharp target. □

Remark 22 (Why the earlier argument lost a logarithm). The multiplicative estimate $\Delta_v(W) \ge D(W)/(\epsilon|W|)$ is sharp when one insists on driving D toward zero at the same density threshold. Here termination is tested at the weaker, original threshold $1 - 1/k$. While that threshold fails, the stronger localized estimate gives unit progress at $\epsilon = 2/k$. Thus the examples with arbitrarily small one-step progress at a fixed threshold do not obstruct this argument.

6 Common sources and the global certificate

For a graph H of mincut $\lambda > 0$, let $s_k(H)$ be the smallest size of a nonempty set R such that

$$F_H(S) \ge \frac{\lambda}{k}|S|, \qquad F_H(V \setminus S) \ge \frac{\lambda}{k}|S| \qquad (S \subseteq V \setminus R). \tag{9}$$

These inequalities are equivalent to existence of two separate feasible flows: one sends λ/k from every outside vertex to R, and the other sends λ/k from R to every outside vertex. For the first equivalence, join a new source to each outside vertex with capacity λ/k, join R to a new sink with capacities exceeding the total demand, and apply max-flow/min-cut [8]. A finite cut avoiding the sink connections has original shore $S \subseteq V \setminus R$; its capacity is at least total demand exactly when the first inequality holds. Saturating all source arcs gives the required supplies. Reverse the graph for the other direction. The two flows need not be feasible when added together; each is certified separately. Such a shared flow into a terminal set is the directed counterpart of the terminal-halving flow of [21], which sends $\lambda/2$ from each discarded terminal to the retained ones.

Corollary 23 (A small common source set)**.** *For the implicit graph H of Lemma 12, with $\lambda > 0$ and integer $1 \le k < n$, one has*

$$s_k(H) = O(n/k).$$

Proof. Let $y, z \le H$ be its in- and out-arborescence packings of value λ, with common root r. Their normalized underlying undirected vectors x^y, x^z are fractional spanning trees. Apply Corollary 21 to both and let R be the union of the resulting deletion sets and $\{r\}$. For every $S \subseteq V \setminus R$,

$$F_H(S) \ge F_y(S) = \lambda(|S| - x^y(E(S))) \ge \frac{\lambda}{k}|S|,$$

and, using the out-packing,

$$F_H(V \setminus S) \ge F_z(V \setminus S) = \lambda(|S| - x^z(E(S))) \ge \frac{\lambda}{k}|S|.$$

Thus R satisfies the two defining inequalities for $s_k(H)$. □

Proposition 24 (Assembly of the certificate)**.** *For the implicit graph H of Lemma 12,*

$$C^{\mathrm{cut}}_{\mathrm{glob}}(G) = O(nk^2 + nk\log^2(2n) + n\,s_k(H)).$$

In particular, $s_k(H) = O(n/k)$ gives a weight-independent $O(n^{5/3})$ certificate. Corollary 23 provides the required source-set bound.

Proof. Use Theorem 14. Certify the two shared divergences associated with a minimizing R at cost $O(n)$ each. A shore avoiding R with at least k vertices then has capacity at least λ; the reverse flow covers shores containing R with complement of size at least k. For cuts splitting R, certify the two value-λ terminal flows between a fixed member of R and every other member, costing $O(n|R|)$. Include an original optimum-shore query. For the concluding bound take $k = \lceil n^{1/3} \rceil$, capped at $n - 1$. □

Proof of Theorem 2. For positive optimum, combine Corollary 23 with Proposition 24. The small-shore certificates, shared flow certificates, and terminal flow certificates all use compiled degrees, pairs, and partial stars of H. Their guarantees therefore hold in every induced H' and hence in every consistent original competitor G'. The original optimum-shore query forces the matching upper bound. Choosing $k = \lceil n^{1/3} \rceil$, capped at $n - 1$, gives $O(n^{5/3} + n^{4/3}\log^2(2n)) = O(n^{5/3})$. For zero optimum, query a zero-capacity nontrivial shore; nonnegative capacities force its optimality in every consistent competitor. □

7 A refinement for bounded symmetrized spread

The proof just completed applies to all nonnegative capacities. On a unit-capacity input, a sharper balance comes from the maximum arc capacity of the pruned graph and its total mass. We first record the two shared star flows and the degree inequality that underlie this balance.

Lemma 25 (Weighted centers)**.** *Suppose every in- and out-degree of G is at least $\lambda > 0$, and its arc capacities are at most M. For every $B \ge M$ there is a nonempty set R of size*

$$O\left(\min\left\{n,\ \frac{n(B + M\log(2 + \lambda/M))}{\lambda}\right\}\right)$$

such that every $v \notin R$ has total capacity at least B to R and at least B from R.

Proof. Put $L = \lambda/M$. If $p = (2B + 16M\ln(2+L))/\lambda \geq 1$, use $R = V$. Otherwise sample each vertex with probability p and add every vertex whose capacity to, or from, the sample is less than B. For each vertex, either sampled star is a sum of independent variables in $[0, M]$, with mean $\eta \geq p\lambda \geq 2B$. The weighted multiplicative Chernoff bound gives

$$\Pr[X < B] \leq \Pr[X < \eta/2] \leq e^{-\eta/(8M)} \leq (2+L)^{-2}.$$

This Chernoff bound follows directly from the usual exponential-moment argument for independent $[0,1]$ variables after dividing by M; integrality is unnecessary. The expected final size is at most $np + 2n(2+L)^{-2}$, which has the claimed order. Some sample attains this bound. If the result were empty, all vertices would have been added, so it is nonempty. Adding deficient vertices only increases the remaining vertices' star capacities. □

For each $v \notin R$, select a total of B units on arcs from v to R, and separately a total of B units on arcs from R to v. These are two feasible flows: distinct outside vertices use distinct tails in the first flow and distinct heads in the second. Their respective divergences are B and $-B$ outside R. They are certified separately, with no simultaneous capacity claim.

Lemma 26 (Covering cuts with a capacity cap)**.** *Suppose the certificate ensures that every competitor has arc capacities at most B, and fixes the original in- and out-degrees. The two preceding flow certificates ensure $F_{G'}(S) \geq \lambda$ for every nontrivial cut that does not split R. Together with root-to-terminal and terminal-to-root flows of value λ for R, they certify global mincut using $O(n|R|)$ additional queries.*

Proof. For $s = |S|$, the degree queries and the cap give

$$F_{G'}(S) \geq s\lambda - Bs(s-1).$$

In particular this is at least λ whenever $1 \leq s \leq \lambda/B$, because $s\lambda - Bs(s-1) - \lambda = (s-1)(\lambda - Bs)$. Singletons have capacity at least λ independently of B. The analogous bound follows from in-degrees for the complement. For a shore avoiding R, the first certified divergence gives $F_{G'}(S) \geq B|S|$, which is at least λ when $|S| \geq \lambda/B$. Thus the two bounds cover every shore avoiding R. For a shore containing R, the second divergence gives $F_{G'}(S) \geq B|V \setminus S|$, and the same argument applies.

Fix $r \in R$. A cut splitting R separates r from another terminal in one direction. Feasible value-λ flows for both ordered pairs exist in the actual graph and are certified by Lemma 6. There are $2(|R|-1)$ such flows in addition to the two shared flows. Finally query an optimum shore. □

Proposition 27 (The refined large case)**.** *For the implicit graph H, the original graph has an unrestricted weighted certificate of size*

$$O\left(n^{3/2} + \frac{n^2 M_H \log(2 + \lambda/M_H)}{\lambda}\right).$$

Proof. Let $k = \lceil\sqrt{n}\rceil$ and $B = \max\{2M_H, \lambda/k\}$. Make the cap queries of Lemma 13 and fix all in- and out-degrees of H. Apply the weighted-center and cut-coverage lemmas to H with target B and actual maximum M_H. The number of centers is

$$O\left(\min\left\{n,\ \frac{n(B + M_H \log(2 + \lambda/M_H))}{\lambda}\right\}\right).$$

Each shared-flow or terminal-flow certificate uses only degree, pair-capacity, and partial-star measurements, so Lemma 12 implements it with $O(n)$ original queries. In particular we do not simulate its written partial-star identity by separately evaluating the two arbitrary cuts of H.

The cap cost is $O(n + n\lambda/B) = O(n^{3/2})$, and

$$n^2 B/\lambda \le 2n^2 M_H/\lambda + n^2/k.$$

The displayed total bound follows. These queries force $\lambda(H') \ge \lambda$, hence also $F_{G'}(S) \ge \lambda$ on every nontrivial shore. One original optimum-shore query gives the matching upper bound; no arbitrary optimum-cut query to H is required. □

Proof of Theorem 4. After the $O(n)$ preprocessing queries, one option is to learn every retained ordinary capacity, making $H' = H$ explicit. Lemma 12 bounds this cost, for every $t > 0$, by

$$O\big(n + q(t) + n\lambda/t\big).$$

Another option is Proposition 27. Since H has mincut λ, $\lambda/M_H \le n-1$, and since $M_H \le U$, the latter cost is at most $O(n^{3/2}+n^2 U \log(2n)/\lambda)$. For nonnegative a, b, c, d, $\min\{a+b, c+d\} \le a+c+\min\{b, d\}$. Taking the geometric mean of the two λ-dependent terms therefore gives

$$O\left(n + q(t) + n^{3/2} + n^{3/2}\sqrt{\frac{U}{t}\log(2n)}\right).$$

For $t \le U$, the square-root term absorbs $n^{3/2} + n$. Take the best threshold and compare with reconstruction. At $t = \nu$ we have $q(t) = 0$, proving the ratio bound.

Finally take the better certificate with Theorem 2, proving (2). □

Proof of Corollary 5. A unit input has every positive symmetrized pair capacity in $\{1, 2\}$. Thus $\kappa \le 2$, and Theorem 4 gives the claim. The zero-optimum case has already been handled by one query. □

An intrinsic parameter, not a uniform logarithmic bound. Cut queries determine u and β but do not determine how each pair capacity is divided between its two directions. For example, on four vertices put $u_e = 1$ on every pair. On the triangle $0, 1, 2$, put capacity $1 - \varepsilon$ clockwise and ε counterclockwise. Put capacities $3/4, 1/4$ on $0 \to 3, 3 \to 0$, respectively, and $1/2$ in both directions on the remaining pairs incident on 3. Then $\beta = (1/2, 0, 0, -1/2)$ and the entire cut function is independent of $0 < \varepsilon < 1/4$, although the ratio of largest to smallest positive directed capacities is $(1 - \varepsilon)/\varepsilon$ and is unbounded. Here $\kappa = 1$ and the input is not Eulerian.

The threshold form is also useful when the symmetrization itself has outliers. If all but k positive pair capacities lie in $[t, U]$, it gives $O(\min\{n^2, k + n^{3/2}\sqrt{(U/t)\log(2n)}\})$. Thus a small number of arbitrarily tiny pairs need not determine the bound. In particular, for unit directed inputs $u_e \in \{0, 1, 2\}$, the $O(n^{3/2}\sqrt{\log(2n)})$ certificate bound is sound against *unrestricted* weighted competitors.

8 Scope and remaining questions

The proof certifies all competitors at once. The orientation masks, branching descriptions, source set, comparison graphs, and tree families are chosen from the actual input, but are thereafter fixed. The certificate queries constrain their interpretations in any competitor. Originally absent pairs may acquire positive capacity. Such pairs carry zero in the transferred signed-flow witness, may become balanced pairs in the cut-equivalent orientation, and are ignored by the implicit pruner. This is why neither a support promise nor an individual-arc oracle is needed.

Certificate selection is the main algorithmic limitation. The proofs use full knowledge of an optimum, extreme packings, and feasible flow witnesses. A short transcript verifying these choices gives no method for finding them with the same number of queries. In particular, conditional procedures supplied with branchings, a lower bound, or a structural template do not establish an unconditional minimum-cut algorithm. We make no such algorithmic claim here.

The structural deletion constant also remains open. The present proof gives at most $\lceil (2n-1)/k \rceil - 2$ deletions for $k \geq 2$. The sharper target

$$|Z| \leq \lceil n/k \rceil - 1$$

would be best possible: let $x_e = 2/n$ on every pair, the average of all spanning stars. An induced set of m vertices has density $(m-1)/n$. Taking the whole remaining set in the desired inequality forces at least $\lceil n/k \rceil - 1$ deletions. This example supplies a lower bound for the deletion problem, not a matching lower bound for directed cut-certificate complexity. Improving that constant alone would not improve the $n^{5/3}$ exponent, whose other costs include nk^2 for fixing and controlling residual capacities.

This manuscript consolidates mathematical proofs from earlier unpublished drafts. It is a draft, not a claim of publication. Appendix A records the correspondence with the current Lean development, including its global certificate endpoints and the differences between its intermediate statements and those here. The full argument is given here apart from the explicitly cited arborescence packing theorem [7, 10] and the max-flow/min-cut theorem [8].

9 The terminal reduction through symmetrization

For completeness, the linear terminal bound can also be derived by a reduction that preserves a specified optimum shore. After $2n$ queries, the imbalance vector β is known. Start with the undirected symmetrization $\mathcal{U}$, add an undirected edge vt of capacity $\beta_v^+ = \max\{\beta_v, 0\}$ and an edge sv of capacity $\beta_v^- = \max\{-\beta_v, 0\}$ for every v, discarding loops and aggregating parallel edges. Call the resulting graph J and put $B = \sum_v \beta_v^-$. For every s–t shore,

$$c_J(S) = c_{\mathcal{U}}(S) + \beta^+(S) + \beta^-(V \setminus S) = 2F_G(S) + B.$$

The extra edges are completely known. An arbitrary cut query to J is simulated by $F_G(S) + F_G(V \setminus S)$ and the known contribution of these extra edges. This is a genuine cut-oracle simulation, unlike the restricted partial-star interface of the implicit pruner.

Here is the needed undirected terminal certificate in the same model. For a maximum signed s–t flow z in J, choose an extreme point of the feasible polytope with its fixed divergence. Its unsaturated positive edges form a forest by the cycle perturbation argument in Lemma 6. Fix those edge capacities and the two saturated partial-star totals at every vertex. On any consistent nonnegative undirected graph, keep the forest flows fixed and saturate every other originally positive pair with its prescribed sign; use zero on originally zero pairs. The divergence is unchanged, so every s–t cut is at least the original flow value. One optimum-cut query fixes a minimizing shore. The undirected partial-star identity

$$c(v, T) = \tfrac{1}{2}\big(c(\{v\}) + c(T) - c(T \cup \{v\})\big)$$

implements all measurements in $O(n)$ undirected queries. Every consistent directed competitor has the same β, and therefore induces a consistent J'. Its terminal cut values obey the same affine identity. The optimum of J', and its certified shore, consequently recover $\lambda_{st}(G') = (\lambda_{st}(J') - B)/2$ and the same shore as for G.

10 The earlier unit-capacity bound and its promise

An earlier unit-capacity construction gives a useful optimum-dependent statement, proved below. Its capacity promise must not be silently transferred to our unrestricted competitor model.

Proposition 28 (Unit inputs with capped competitors)**.** *Let G have unit capacities and $\lambda > 0$. Against competitors with $a'_{uv} \in [0,1]$ on every ordered pair, it has a certificate of size*

$$O\left(\min\left\{n\lambda, \frac{n^2 \log(2\lambda)}{\lambda}\right\}\right).$$

The $O(n\lambda)$ alternative remains valid against all nonnegative weighted competitors.

Proof. Integral arborescence packing [7] gives λ arc-disjoint out-arborescences and, separately, λ arc-disjoint in-arborescences about a common root. Their union has at most $2\lambda(n-1)$ arcs and every nontrivial outgoing cut has at least λ arcs. The two families may overlap; disjointness is required only within each family. Lemma 7 and an optimum-shore query give the first bound against unrestricted competitors.

For the other bound apply Lemma 25 directly to G with $M = B = 1$. It gives a nonempty set $|R| = O(n \log(2+\lambda)/\lambda)$, because every in- and out-degree is at least λ. Use the two shared star-flow certificates and the root-to-terminal certificates from Lemma 26. The competitor promise supplies the arc cap needed in its degree inequality, so no cap-verification queries are necessary. Their cost is $O(n|R|) = O(n^2 \log(2\lambda)/\lambda)$; here $\lambda \geq 1$. Include an optimum shore and take the smaller certificate. □

Taking the geometric mean and using $1 \leq \lambda \leq n-1$ yields $O(n^{3/2}\sqrt{\log(2n)})$ under the stated cap. The main paper's Corollary 5 obtains the same uniform order against unrestricted competitors by the additional pruning and mass-charged cap verification. It does not assert that the optimum-dependent second term above remains valid without its promise.

The need for the promise in the earlier construction is concrete. Take the complete bidirected graph on $\{r, a, b, c, d\}$, with $\lambda = 4$, and use $R = \{r\}$ in the shared-star construction. Keep both arcs of every root spoke at capacity one. Give capacity three to both directions of ab and cd, and zero to all other leaf-to-leaf arcs. This weighted competitor has the same in- and out-degrees and the same symmetrized root spokes and associated star totals as the input, but $F(\{a,b\}) = 2$. Including a singleton optimum-shore query does not resolve the ambiguity, since its value is still four. A degree argument with an unverified cap of one would therefore be unsound.

AI use

The results in this paper were obtained by prompting Codex GPT-6 Astra (OpenAI), which also wrote this preliminary draft. The Lean proofs (Appendix A) were developed by Claude Opus 5.5 (Anthropic) and audited by GPT-6 Astra.

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A Correspondence with the Lean formalization

This appendix relates all numbered definitions, lemmas, theorems, corollaries, and propositions in the paper to the Lean library https://github.com/troyjlee/cut-query-complexity, version 0.1.0. The tables give declaration names, source modules, and the mathematical content of the Lean statements. A correspondence may consist of several declarations. Where the development proves a different statement or uses a different route, this is stated explicitly; a related declaration is not a claim that the paper's full statement has been formalized.

All 83 referenced declarations were checked against a fresh build of version 0.1.0 with `#check` and `#print axioms`. The latter reported only `propext`, `Classical.choice`, and `Quot.sound`, or no axioms.

Declaration names below use three namespace abbreviations:

`DC`	`CutQueryComplexity.DirectedCertificate`
`C`	`CutQueryComplexity.DirectedCertificate.Completion`
`FF`	`CutQueryComplexity.FractionalForest`

All file paths are relative to `CutQueryComplexity/`; except where a path starts with `Graph/` or `FractionalForest/`, its prefix is `DirectedCertificate/`.

The formal graph type is `DC.WeightedDiGraph V`, where V is a finite vertex type with decidable equality. It includes real-valued capacities, their nonnegativity, and a zero diagonal. The expressions `G.out S`, `G.minValue`, `G.sym.w u v`, and `G.imbalance v` represent $F_G(S)$, $\lambda(G)$, $u_{\{u,v\}}$, and β_v, respectively. Query families and shores are finite sets. In each certificate theorem the queries and witness shore are chosen before the universal quantifier over consistent competitors. The competitor type imposes no capacity cap or support restriction, including in the unit-input theorems. All costs count original outgoing-cut queries.

For the explicit budgets below, put

$$n = |V|, \qquad L = \lceil \log_2(2n) \rceil, \qquad A = \lceil n/k \rceil.$$

These are represented by `DC.logL` and `DC.ceilDiv`; `DC.budget` gives

$$B_c(n,k) = c(nk^2 + nkL^2 + n(A+1)).$$

The symbol C_{w} below denotes the fixed real constant `C.weightedHeadlineConstant`. The formal definitions `C.symMaximum`, `C.symMinimumPositive`, `C.symSpread`, and `DC.PruningData.q` are exactly $U, \nu, \kappa, q(t)$. Their positivity and input-capacity bounds are proved by `C.intrinsic_parameters` in `Completion/Corollaries.lean`.

For the unnumbered definitions in Section 5, `FF.mass` is $x(E(S))$ and `FF.IsFractionalForest` is the fractional-forest predicate. Both are in `Graph/FractionalForest.lean`; the mass is half the symmetric double sum, so each unordered pair is counted once. The functions g, D, Δ are `FF.gfun`, `FF.Dpot`, and `FF.gain` in `FractionalForest/Potential.lean`. The density ρ and the predicate that K is densest in W are `FF.dens` and `FF.IsDensest` in `FractionalForest/Progress.lean`.

A.1 Certificate definitions and main results

Paper statement	Lean declarations and their statements
Definition 1 Witness certificate	`DC.Consistent`, `DC.IsMinCutWitnessCertificate`, `DC.IsMinCutWitnessCertificate.isValueCertificate` (`Basic.lean`); `C.IsTerminalWitnessCertificate`, `C.IsTerminalWitnessCertificate.values` (`Completion/Terminal.lean`). The global predicate says $S_* \in Q$ and that S_* is a proper minimum shore of every graph agreeing on Q. The value-certificate theorem supplies equality of minimum values. The terminal predicate and its value theorem give the corresponding s–t statements.
Theorem 2 Arbitrary-weight global certificates	`C.exists_uniform_certificate`, `C.uniform_certificate_target` (`Completion/Main.lean`) give $\lvert Q\rvert \le B_{512}(n,k)$ for $1 \le k < n$, with a queried optimum shore valid in every competitor. `C.exists_weighted_global_witnessCertificate` (`Completion/Corollaries.lean`) gives $\lvert Q\rvert \le C_{\mathrm{w}} n^{5/3}$ for $n \ge 2$. `DC.exists_zero_witnessCertificate` (`Basic.lean`) is the one-query case $\lambda = 0$. The ceiling term $n(A+1)$ is $O(n^2/k)$ in the stated range.
Theorem 3 Terminal certificates	`C.exists_exact_terminal_certificate`, `C.exists_zero_terminal_certificate` (`Completion/Terminal.lean`). For $s \ne t$, at most $18n+1$ queries determine a queried optimum s–t shore and its exact value in every nonnegative weighted competitor. If the terminal value is zero, the optimum-shore query alone suffices.
Theorem 4 Spread-sensitive refinement	`C.exists_intrinsic_paper_certificate`, `C.exists_intrinsic_ratio_paper_certificate` (`Completion/Corollaries.lean`). For $n \ge 2$ and $\lambda > 0$, a witness certificate has size at most the minimum of $C_{\mathrm{w}} n^{5/3}$ and $\inf_{0<t\le U} 1000[q(t) + n^{3/2}\sqrt{(U/t)\log(2n)}]$; the ratio version substitutes $1000n^{3/2}\sqrt{\kappa \log(2n)}$. In the same module, `C.exists_intrinsic_inf_certificate` and `C.exists_intrinsic_ratio_certificate` prove the stronger bounds with the factor $\log(2n)$ omitted from the square roots.
Corollary 5 Unit-capacity global certificates	`C.exists_unit_paper_certificate` (`Completion/Corollaries.lean`) gives $\lvert Q\rvert \le 600n^{3/2}\sqrt{\log(2n)}$ for binary input capacities. `C.exists_unit_global_certificate` (`Completion/Main.lean`) proves the stronger $\lvert Q\rvert \le 600n^{3/2}$. Both use the unrestricted weighted witness predicate; no unit-capacity condition is imposed on competitors.

A.2 Flow certificates, pruning, and small shores

Paper statement	Lean declarations and their statements
Lemma 6 Arbitrary real flow divergences	`DC.exists_flowCert_directed` (`FlowCert.lean`). For any feasible directed flow f, there is a family of at most $18n$ queries such that every consistent G' satisfies $\sum_{v\in S} \mathrm{netOut}(f,v) \le F_{G'}(S)$ for all S. `DC.exists_implicitFlowCert` (`FlowMeasure.lean`) gives the same guarantee for an implicit graph using at most $5cn$ queries, given its `DC.ImplicitMeasures` interface of cost c.
Lemma 7 Sparse weighted domination	No general certificate theorem for an arbitrary explicit $H \le G$ with s positive arcs was located. `DC.PruningCert.out_le` (`Pruning.lean`) certifies domination for the particular implicit pruner. `C.exists_retained_witnessCertificate` (`Completion/Spread.lean`) fixes that pruner's ordinary retained arcs and gives a global witness certificate of size $30n + 3\lvert\mathrm{ret}\rvert + 1$. This is a specialized replacement in the endpoint proof, not the general statement of the lemma.

Paper statement	Lean declarations and their statements (continued)
Lemma 8 Real-capacity branching packing	`DC.exists_outArb_packing`, `DC.exists_inArb_packing` (`Graph/ArborescencePacking.lean`). The incoming, respectively outgoing, cut condition away from r gives nonnegative coefficients on a finite set of rooted trees, summing to α, with each arc load bounded by its real capacity. Both statements allow $\alpha = 0$. Out-trees are represented by reversing the arcs of the same `DC.InArb` index type.
Lemma 9 Sparse weighted spanning certificate	`DC.exists_sparse_inArb_packing`, `DC.exists_sparse_outArb_packing` (`SparseBranching.lean`). Each packing has value λ, total load $\lambda(n-1)$, and at most $2(n-1)$ arcs with load strictly between zero and their capacity. Thus the formal free-arc bound is one larger per packing than the paper's $2n-3$. The combined support bound $2(n-1)\lambda/\mu + 4n - 6$ is not a stated Lean theorem; the two packing statements support its version with $4(n-1)$ instead.
Corollary 10 Minimum-positive-weight bound	`C.exists_min_positive_weight_certificate` (`Completion/Corollaries.lean`). For $n \geq 2$ and any $\mu > 0$ bounding every positive input arc capacity from below, there is an unrestricted witness certificate with $\lvert Q\rvert \leq 60n + 6n\lambda/\mu + 1$. This conclusion is obtained through the pruner's retained-support bound, without using a general sparse-domination lemma.
Lemma 11 Cut-equivalent orientation	`DC.exists_orientationCert`, `DC.OrientationCert`, `DC.OrientationData.toGraph_ord`, `DC.OrientationData.toGraph_c_zero` (`Orientation.lean`). The package uses at most $18n$ queries and at most n exceptional pairs, and gives exact cut-function equality for every consistent competitor. The paper states the sharper exceptional-pair count $n-1$. Ordinary originally positive pairs have one prescribed direction; originally zero pairs may acquire equal capacity in both directions.
Lemma 12 Implicit graph of small total capacity	`DC.exists_pruningCert`, `DC.PruningCert` (`Pruning.lean`); `C.pruned_arc_le_pair_bound` (`Completion/Spread.lean`). The package has at most $30n$ queries and $5n$ exceptional pairs. It gives total capacity at most $2(n-1)\lambda$, the same optimum value, competitor-wise cut domination, and stars of cost at most three original queries. Degrees and pairs are special cases. Its fields bound each arc by λ and the support by $10n + q(t) + 2(n-1)\lambda/t$ for $t > 0$; the last declaration bounds each arc by any upper bound U on original pair capacities.
Lemma 13 Cap verification	`C.exists_pruning_mass_cap` (`Completion/MassCap.lean`). If $B > 0$ and every arc of the actual pruner has capacity at most $B/2$, at most $3n + 12(n-1)\lambda/B$ queries force every induced competitor arc to have capacity at most B. These are additional queries to the pruner's initial transcript.
Theorem 14 All small shores	`C.exists_smallShore_certificate` (`Completion/Main.lean`). For a chosen vertex r and $1 \leq k < n$, at most $400nk^2 + 60nkL^2$ queries force $F_{G'}(S) \geq \lambda(G)$ for every proper shore with $\lvert S\rvert \leq k$ or $\lvert S^c\rvert \leq k$. The statement constructs the queries before quantifying over all weighted competitors and includes the case $\lambda(G) = 0$.
Lemma 15 Connected shores of a tree	`C.exists_connected_tree_certificate` (`Completion/Main.lean`); `C.ParentConnected` (`Completion/Centroid.lean`). Given an implicit measurement interface of cost c, a lower bound $\lambda \geq 0$ on all proper input cuts, any rooted tree T, and a vertex r, at most $5cn\lceil \log_2 n \rceil$ queries certify the lower bound for every nonempty T-connected shore avoiding r. There is no hypothesis that tree edges belong to the graph.

A.3 Fractional forests and the common-source construction

Paper statement	Lean declarations and their statements
Lemma 16 Localization to maximum density	`FF.densest_core`, `FF.gain_mono`, `FF.gfun_le_h_mul` (`FractionalForest/Progress.lean`). For a fractional forest and densest $K \subseteq W$ with $D(W) > 0$, the first statement gives $h > 0$, $1 - \rho > 0$, $D(K) = g(K) = h\lvert K\rvert$, and $D(W)/\lvert W\rvert \le h$. The second gives $\Delta_v(K) \le \Delta_v(W)$ for $v \in K$. The last gives $g(S) \le h\lvert S\rvert$ for every $S \subseteq W$; applying it to $K \setminus T$ yields the paper's loss inequality.
Lemma 17 Minimal rooted-forest covers	`FF.RootedForest.rooted_cover`, `FF.RootedForest.rooted_cover_weighted` (`FractionalForest/RootedCover.lean`). For a cover indexed by a finite type of size m, with a private vertex for each member, the sum of entering-arc counts and root counts is at least m. Weighting the root counts by $\gamma > 0$ gives at least $m \min\{1, \gamma\}$. The private-vertex condition expresses exactly that no member of the cover can be removed.
Corollary 18 Concentration of fractional roots	`FF.root_concentration` (`FractionalForest/RootedCover.lean`). For a finite mixture of rooted forests on a nonempty vertex type, with nonnegative coefficients summing to one, and $\gamma > 0$, there is v such that every $T \ni v$ satisfies $\min\{1, \gamma\} \le y(\delta^-(T)) + \gamma r(T)$. The expected arcs and root marginals are `FF.arcVec` and `FF.rootVec`.
Lemma 19 Uniform roots on a densest set	`FF.uniform_roots` (`FractionalForest/Progress.lean`). For a fractional forest on a nonempty finite type K, with $x(E(S)) \le \rho\lvert S\rvert$ for every S and $x(E(K)) = \rho\lvert K\rvert$, there is a nonnegative mixture of rooted forests of total weight one. Its expected arcs satisfy $y_{uv} + y_{vu} = x_{\{u,v\}}$, every out-degree is ρ, and every root marginal is $1 - \rho$. The mixture is indexed by arborescences on K with one additional vertex, whose removal gives the rooted forests.
Theorem 20 Deletion progress	`FF.deletion_progress`, `FF.exists_gain_ge_div` (`FractionalForest/Progress.lean`). The first gives $v \in K$ with $\Delta_v(W) \ge \Delta_v(K) \ge \min\{1, h/(1-\rho)\}$ for a densest K with $h > 0$. The second gives $v \in W$ with $\Delta_v(W) \ge D(W)/(\epsilon\lvert W\rvert)$ whenever $D(W) > 0$. The numerical comparison with h/ϵ is used in its proof. `FF.unit_progress` proves the unit-gain version at slack $2/k$ whenever $k \ge 3$ and the target density is violated.
Corollary 21 Structural density deletion	`FF.forest_deletion_main`, `FF.forest_deletion_uniform_main` (`SourceSetMain.lean`); `FF.budgetQ_eq_ceil`, `FF.budgetQ_le` (`FractionalForest/Deletion.lean`). For $1 \le k \le n$, the first theorem produces Z with all remaining induced densities at most $1 - 1/k$, using at most $n - 1$ deletions for $k = 1$ and `FF.budgetQ n k` otherwise. The budget lemmas identify this with $\lceil (2n-1)/k \rceil - 2 \le 2(\lceil n/k \rceil - 1)$ for $k \ge 2$. The `main` theorems discharge the progress hypothesis present in the earlier deletion module.
Corollary 23 Small common source set	`DC.small_source_set_main`, `DC.small_source_set_flows_main` (`SourceSetMain.lean`). Given in- and out-arborescence packings of value $\lambda > 0$ about r, for $1 \le k \le n$ these produce $r \in R$ with $\lvert R\rvert \le 4(\lceil n/k \rceil - 1) + 1$. For $S \cap R = \varnothing$, both $F_H(S)$ and $F_H(S^c)$ are at least $(\lambda/k)\lvert S\rvert$. The flow version gives separate feasible flows with divergences λ/k and $-\lambda/k$ outside R. Packing existence is supplied by Lemma 8's Lean counterparts.
Proposition 24 Assembly of the certificate	`C.exists_source_route_certificate` (`Completion/SourceRoute.lean`) accepts a source set R and its two feasible flows, and uses at most $10cn(\lvert R\rvert + 1)$ queries to cover all shores with $\lvert S\rvert, \lvert S^c\rvert \ge k$ through an implicit interface of cost c. `C.exists_uniform_certificate` (`Completion/Main.lean`) combines the constructed source set with the small-shore certificate and an optimum-shore query. The optimization parameter $s_k(H)$ is not a separate Lean definition; its role is represented by the supplied R and flows.

A.4 Spread refinement and the earlier unit bound

Paper statement	Lean declarations and their statements
Lemma 25 Weighted centers	No direct counterpart of the sampled-center statement was located. The formal spread refinement uses the common-source construction instead, as recorded in `Completion/Spread.lean`. Its sources route flow along paths and require cut lower bounds; they do not assert the star-capacity conclusion of this lemma under degree lower bounds alone.
Lemma 26 Covering cuts with a cap	`C.degree_cap_cut_lower_bound` (`Completion/SmallShore.lean`) gives $F(S) \geq \lambda$ from out-degrees at least λ, arc cap B, $S \neq \varnothing$, and $B\|S\| \leq \lambda$. `C.sum_netOut_supply`, `C.sum_netOut_demand`, and `C.exists_root_split_certificate` (`Completion/SourceRoute.lean`) give the two shared-flow divergence identities and the $10cn\|R\|$ bound for cuts splitting R. Together with the flow compiler these are components of the paper's proof; the arbitrary-B cap-and-star-flow assembly is not exported as one theorem.
Proposition 27 Refined large case	`C.exists_mass_large_paper_certificate` (`Completion/Corollaries.lean`) gives at most $1200[n^{3/2} + n^2 M \log(2 + \lambda/M)/\lambda]$ queries for any $M > 0$ bounding the pruner's arc capacities, when $n \geq 2$ and $\lambda > 0$. `C.exists_pruned_refined_large_certificate` (`Completion/Spread.lean`) proves the stronger $600[n^{3/2} + n^2 M/\lambda]$ bound using common sources. Both certify a queried optimum shore against unrestricted weighted competitors.
Proposition 28 Unit inputs with capped competitors	`C.exists_unit_mass_certificate` (`Completion/Main.lean`) gives an unrestricted witness certificate with at most $30n + 6n\lambda + 1$ queries for a unit input and its pruner. Since a positive unit optimum is at least one, this gives the $O(n\lambda)$ alternative. No statement of the capped-competitor alternative $O(n^2 \log(2\lambda)/\lambda)$, or of the displayed minimum of the two bounds, was located. The stronger uniform unit theorem in the first table does not by itself imply that optimum-dependent alternative.

The development therefore contains the global and terminal endpoints, the small-shore construction, and the fractional-forest deletion argument, while some auxiliary claims above are represented only by components or by an alternative proof. Section 9 contains no numbered statement; the terminal endpoint listed above uses the direct flow certificate, rather than that section's alternative reduction through an undirected terminal instance.